

Stony Brook University
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Spring 2018

ASSIGNMENT 5

Due: May 2, 2018, in class

INSTRUCTIONS: Show all your work to get full points. Please also cut and paste at the end of your submission the R code you used. The last problem is optional, if you attempt it you can get extra credit.

1. You have a data set consisting on a cross-sample of 4131 high-school students. The variables you have are: *sat*, which is the combined SAT score; *hsize* is the size of the student's high school graduating class, in hundreds; *female* is a gender dummy variable, having a value of one if female or zero if male; and *black* which is a race dummy variable equal to one for blacks and zero otherwise.

The following models are estimated.

	<i>Dependent variable:</i>	
	<i>sat</i>	
	(1)	(2)
<i>hsize</i>	19.297*** (3.817)	3.933*** (1.178)
<i>hsize</i> ²	-2.195*** (0.524)	
<i>hsize</i> ² - 7 <i>hsize</i>		-2.195*** (0.524)
<i>female</i>	-45.091*** (4.230)	-45.091*** (4.230)
<i>black</i>	-169.813*** (15.434)	-169.813*** (15.434)
<i>female</i> × <i>black</i>	62.306*** (19.320)	62.306*** (19.320)
Constant	1,028.097*** (6.360)	1,028.097*** (6.360)
Observations	4,131	4,131
<i>R</i> ²	0.086	0.086
Adjusted <i>R</i> ²	0.085	0.085
Residual Std. Error	133.369	133.369
Overall F Statistic (df = 5; 4131)	77.524***	77.524***
<i>F</i> Statistic <i>black</i> + <i>female</i> × <i>black</i> = 0 (df=1;4131)	84.965***	
Note: *p<0.1; **p<0.05; ***p<0.01		

- (a) Consider the model in column (1). Write down the population model that the researcher has in mind.
- (b) Is there strong evidence that $hsize^2$ should be included in the model?
- (c) Consider student A , which graduated from a high school class of 300 ($hsize = 3$), is female and not black. Student B graduated from a high school class of 400 ($hsize = 4$) and is also female and not black. Let $\Delta sat = sat_B - sat_A$ denote the difference in SAT scores where sat_B and sat_A are the SAT scores of students B and A respectively. Rewrite the regression model so that you can construct a 95% confidence interval around Δsat . Compute this confidence interval. *Hint: You will have to use the regression in model (2). How do you rewrite the regression model so that you can use it?*
- (d) Holding the value of $hsize$ fixed, what is the estimated difference in SAT score between black females and nonblack females? State the null hypothesis that you would need to test to determine if the difference is statistically significant. Test it at the 5% level. After you test this null hypothesis, construct a 95% confidence interval around the estimated difference. *Hint: Look at the F statistic at the bottom of the table. What is its meaning? To construct the confidence interval, you will have to use one of the formulas we saw in class.*
2. This problem is inspired by a study of the “gender gap” in earnings in top corporate jobs (Bertrand, M. and Hallock, K.F., 2001. *The gender gap in top corporate jobs. Industrial & Labor Relations Review*, 55(1), pp.3–21). The study compares total compensation among top executives in a large set of U.S. public corporations in the 1990s.

- (a) Let $Female$ be an indicator variable that is equal to 1 for females and 0 for males. A regression of the logarithm of earnings onto $Female$ yields

$$\widehat{\ln(Earnings)} = \underset{(0.01)}{6.48} - \underset{(0.05)}{0.44} Female, SER = 2.65.$$

- The estimated coefficient on $Female$ is -0.44 . Explain what this value means.
 - The SER is 2.65. Explain what this value means.
 - Does this regression suggest that female top executives earn less than top male executives? Explain.
 - Does this regression suggest that there is gender discrimination?
- (b) Two new variables, the market value of the firm (a measure of firm size, in millions of dollars) and stock return (a measure of firm performance, in percentage points), are added to the regression:

$$\widehat{\ln(Earnings)} = \underset{(0.03)}{3.86} - \underset{(0.04)}{0.28} Female + \underset{(0.004)}{0.37} \ln(MarketValue) + \underset{(0.003)}{0.004} Return, n = 46670, \bar{R}^2 = 0.345$$

- The coefficient on $\ln(MarketValue)$ is 0.37. Explain what this value means.
- (c) Are large firms more likely than small firms to have female top executives? Explain.
3. Consider the regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{1i} \times X_{2i} + u_i.$$

Calculate

- $\frac{\Delta Y_i}{\Delta X_1}$ (effect of a change in X_1 holding X_2 constant).
- $\frac{\Delta Y_i}{\Delta X_2}$ (effect of a change in X_2 holding X_1 constant).

- (c) If X_1 changes by ΔX_1 and X_2 changes by ΔX_2 , calculate ΔY .
4. Use the data in BWGHT2 for this exercise.

- (a) Estimate the equation

$$\log(bwght) = \beta_0 + \beta_1 npvis + \beta_2 npvis^2 + u$$

by OLS, and report the results in the usual way. Is the quadratic term significant?

- (b) Show that, based on the equation from part (a), the number of prenatal visits that maximizes $\log(bwght)$ is estimated to be about 22. How many women have at least 22 prenatal visits in the sample?
- (c) Does it make sense that birth weight is actually predicted to decline after 22 prenatal visits? Explain.
- (d) Add mother's age to the equation, using a quadratic functional form. Holding $npvis$ fixed, at what mother's age is the birth weight of the child maximized? What fraction of women in the sample are older than the "optimal" age? *Hint: the quadratic equation $y = A + Bx + Cx^2$ defines a parabola, where A , B and C are constants. If $C < 0$ then the parabola has a global maximum at $x^* = -\frac{B}{2C}$.*
- (e) Would you say that mother's age and number of prenatal visits explain a lot of the variation in $\log(bwght)$?
- (f) Using quadratics for both $npvis$ and age , decide whether using the natural log or the level of $bwght$ is better for predicting $bwght$.
5. (Extra Credit) Consider a model where the return to education depends upon the amount of work experience (and viceversa):

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 educ \times exper + u$$

- (a) Show that the return to another year of education (in decimal form), holding $exper$ fixed is $\beta_1 + \beta_3 exper$.
- (b) State the null hypothesis that the return to education does not depend on the level of $exper$. What do you think is the appropriate alternative?
- (c) Use the data in WAGE2 to test the null hypothesis in (b) against your stated alternative.
- (d) Let θ_1 denote the return to education (in decimal form), when $exper = 10$: $\theta_1 = \beta_1 + 10\beta_3$. Obtain $\hat{\theta}_1$ and a 95% confidence interval for θ_1 . *Hint: Rewrite the regression. Here is a useful trick to do it: write $\beta_1 = \theta_1 - 10\beta_3$ and plug this into the equation; then rearrange.*